Kejing Xia

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EDUCATION

Georgia Institute of Technology **Atlanta, GA**

- **Ph.D. in** Computer Science, **advised by Wenke Lee** Aug. 2025 – Present

Wuhan University **Wuhan**

- **B.Eng. in** Communications Engineering (Artificial Intelligence Track) Sept. 2021 – Jun. 2025
- **GPA:** 3.90/4.00 | **Rank:** Top 5% | **1st class Academic Scholarship** & **Merit Students Award**

RESEARCH EXPERIENCE (Project leader & Core member)

Georgia Institute of Technology, GTISC Lab. | Ph.D. Student Jan. 2026 – Present

MetaState: Persistent Working Memory for Discrete Diffusion Language Models

- Developed MetaState, a lightweight recurrent **working-memory augmentation** for **dLLMs** that mitigates the **Information Island bottleneck** by persisting **cross-step continuous information**. Yielding consistent gains on reasoning/coding benchmarks with **negligible added parameters**.

EDITOR: Effective and Interpretable Prompt Inversion for Text-to-Image Diffusion Models

- Developed EDITOR, an interpretable **prompt inversion** framework for text-to-image diffusion models that introduces **continuous contextual-latent optimization** via caption initialization, **latent space reverse-engineering**, and embedding-to-text converting.

Georgia Institute of Technology, EIC Lab. | Undergrad to Ph.D. Student May. 2024 – Feb. 2026

R^2-dLLM: Accelerating Diffusion Large Language Models via Spatio-Temporal Redundancy Reduction

- Developed R^2-dLLM, a unified **redundancy reduction** framework to accelerate **dLLM** decoding. Identified **spatial redundancy** (confidence clusters, positional uncertainty) and **temporal redundancy** (early-converged tokens). Proposed 1) training-free decoding mechanism, and 2) redundancy-aware **SFT** to align efficient trajectories.

Soft Token Interpolation for Masked Diffusion LLM: Training-Free Step-to-Step State for dLLM Decoding

- Proposed a training-free **interpolation** for **dLLMs**: cache per-token top-k candidate probabilities and reinject them as **probability-weighted embeddings mixtures** to introduce **a continuous intermediate state** between mask and decoded tokens, mitigating **hard-mask information loss** and enabling **step-to-step semantic carryover**.

LongMamba: Enhancing Mamba's Long-Context Capabilities via Training-Free Receptive Field Enlargement

- Discovered **local vs. global channels** in **Mamba LLMs** via per-channel receptive-field analysis; identified **exponential hidden-state decay** in **global channels** as the key bottleneck for **long-context extrapolation**.
- Proposed a training-free receptive-field enlargement pipeline: **classify channels** by cumulative decay and apply **token filtering in global channels** to prevent unimportant-token accumulation and **maintain long-range memory**.

LAMB: A Training-Free Method to Enhance the Long-Context Understanding of SSMs via Attention-Guided Token Filtering

- **Analyzed "attention" patterns** of SSMs and proposed a **pseudo-attention importance metric** to preserve critical tokens during inference, enabling **training-free long-context extension** for **Mamba & hybrid LLM** models.

Wuhan University, Signal Processing Lab. | Undergrad Student Oct. 2023 – Apr. 2024

BRS-E^2NeRF: Event-Enhanced Neural Radiation Fields from Blurry Rolling Shutter Images

- Developed a **unified event+RGB pipeline** for **NeRF under fast motion** that **jointly corrects rolling-shutter distortion and motion blur** from blurry RS images for sharp **3D reconstruction** and **novel-view synthesis**.

PROFESSIONAL EXPERIENCE

Bosch Research & Innovation Software Development Co., Ltd | Lead Research Intern Jul. 2024 – May. 2025

D^2GS: Dense Depth Regularization for LiDAR-free Urban Scene Reconstruction

- Co-optimized **diffusion-based depth generation** and **3D Gaussian Splatting** via an **iterative enhancement training** pipeline for LiDAR-free **urban reconstruction**; Added a **Gaussian Scene Graph** with a Road Node and road-specific losses; conducted rigorous benchmarking on **Waymo NOTR Dynamic32**.

RESEARCH INTERESTS

- **Diffusion LLM**: Decoding Acceleration, Hybrid-continuous dLLM.
- **LLM Reasoning**: Latent reasoning, LLM alignment.
- **Efficient LLMs**: Long-context modeling, Training-free optimization.

- **3D Vision**: 3D Gaussian Splatting, Event-based vision, Neural Radiance Fields.

PUBLICATION (*: CO-FIRST)

[1] **Zhifan Ye*, Kejing Xia***, Yonggan Fu, Xin Dong, Jihoon Hong, Xiangchi Yuan, Shizhe Diao, Jan Kautz, Pavlo Molchanov, Yingyan (Celine) Lin, *"LongMamba: Enhancing Mamba's Long-Context Capabilities via Training-Free Receptive Field Enlargement", The Thirteenth International Conference on Learning Representations* **(ICLR 2025)**

[2] Zhifan Ye, Zheng Wang, **Kejing Xia**, Jihoon Hong, Leshu Li, Lexington Whalen, Cheng Wan, Yonggan Fu, Yingyan (Celine) Lin, Souvik Kundu, *"LAMB: A Training-Free Method to Enhance the Long-Context Understanding of SSMs via Attention-Guided Token Filtering", The 62nd Annual Meeting of the Association for Computational Linguistics* **(ACL 2025)**

[3] **Kejing Xia**, Jidong Jia, Ke Jin, Yucai Bai, Li Sun, Dacheng Tao, Youjian Zhang, *"D^2GS: Dense Depth Regularization for LiDAR-free Urban Scene Reconstruction", The Thirty-ninth Annual Conference on Neural Information Processing Systems* **(NeurIPS 2025)**

[4] **Huijiao Wang*, Kejing Xia***, Yangguang Wang, Chi Zhang, Mingyuan Lin, Lei Yu, *"BRS-E^2NeRF: Event Enhanced Neural Radiance Fields from Blurry Rolling Shutter Images",* ***The Journal of Pattern Recognition***

[5] **Kejing Xia**, Lixuan Wei, Lei Yu, *"A Spatio-temporal Event Transformer on Versatile Tasks for Human Behavior Analysis", MiGA at The 33rd International Joint Conference on Artificial Intelligence* (**IJCAI-W 2024)**